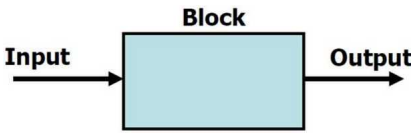


Lesson 2 Outline

- Block Diagram fundamentals
- Block Diagram of a continuous (analog) control system
- Terminology of the Closed-Loop System Block Diagram
- Block Diagram of Discrete-time components
- Control Systems and Computer controlled systems

A block diagram is a pictorial representation of the cause-and-effect relationship between the input and output of a physical system.



It provides a convenient and useful method in characterizing the functional relationships among the various components of the control system. System components are alternatively called elements of the system.

The block usually contains a description of or the name of the element, or the symbol for the mathematical operation to be performed on the input to yield the output. The arrow represents the direction of information or signal flow.

Operations like addition or subtraction have a different special representation that looks like a small circle referred to as a summing point with the appropriate plus or minus sign associated with the arrows entering the circle.

The output is the algebraic sum of the inputs in which any number of inputs may enter a summing point.

In order to have the same signal or variable be an input to more than one block or summing point, a take-off point is used. This permits the signal to be unaltered along several different paths to several destinations.

Block diagram of a continuous (analog) feedback control system

The blocks representing the various components of a system are connected in a fashion which characterizes their functional relationships within the system. The basic configuration of a simple closed-loop feedback system with a single input and a single output is referred as SISO.

Block diagram of a continuous (analog) feedback control system

Terminology of a closed-loop block diagram system Definition 2.1: The plant (or process, or controlled system) g2 is the system,

subsystem, process, or object controlled by the feedback control system. Definition 2.2: The controlled output c is the output variable of the plant, under the control of the feedback control system.

Definition 2.3: The forward path is the transmission path from the summing point to the controlled output c.

Terminology of a closed-loop block diagram system Definition 2.4: The feedforward (control) elements g1 are the components of

the forward path that generate the control signal u or m applied to the plant. Note: Feedforward elements typically include controller(s), compensator(s) (or equalization elements), and/or amplifiers.

Definition 2.5: The control signal u (or manipulated variable m) is the output signal of the feedforward elements g1 applied as input to the plant g2.

Terminology of a closed-loop block diagram system

Definition 2.6: The feedback path is the transmission path from the controlled output c back to the summing point.

Definition 2.7: The feedback elements h establish the functional relationship between the controlled output c and the primary feedback signal b. Note: Feedback elements typically include sensors of the controlled output c, compensators, and/or controller elements.

Terminology of a closed-loop block diagram system

Definition 2.8: The reference input r is an external signal applied to the feedback control system, usually at the first summing point, in order to command a specified action of the plant. It usually represents ideal (or desired) plant output behavior.

Definition 2.9: The primary feedback signal b is a function of the controlled output c, algebraically summed with the reference input r to obtain the actuating (error) signal e, that is, $r \pm b = e$.

Note: An open-loop system has no primary feedback signal.

Terminology of a closed-loop block diagram system Definition 2.10: The actuating (or error) signal is the reference input signal r plus or minus the primary feedback signal b. The control action is generated by the actuating (error) signal in a feedback control system.

Note: In an open-loop system, which has no feedback, the actuating signal is equal to r.

Definition 2.11: Negative feedback means the summing point is a subtractor, that is, $e = r - b$. Positive feedback means the summing point is an adder, that is, $e = r + b$.

Block diagrams of Discrete-Time components, Control Systems, and Computer-Controlled Systems

A discrete-time (sampled-data or digital) control system was defined in

Definition 1.11 as one having discrete-time signals or components at one or more points in the system.

EXAMPLE: A digital computer or microprocessor is a discrete-time (discrete or digital) device, a common component in digital control systems. The internal and external signals of a digital computer are typically discrete-time or digitally coded.

EXAMPLE: A discrete system component (or components) with discrete-time input u(tk) and discrete-time output y(tk) signals, where tk are discrete instants of time, k = 1,2,..., etc., may be represented by a block diagram, below.

Many digital control systems contain both continuous and discrete components. One or more devices known as samplers, and others known as holds, are usually included in such systems.

Definition 2.12: A sampler is a device that converts a continuous-time signal, say u(t), into a discrete-time signal, denoted u*(t), consisting of a sequence of values of the signal at the instants t1, t2,..., that is, u(t1), u(t2),..., etc.

Ideal samplers are usually represented schematically by a switch, as shown in figure below, where the switch is normally open except at the t1, t2,...,etc., when it is closed for an instant. The switch also may be represented as enclosed in a block.

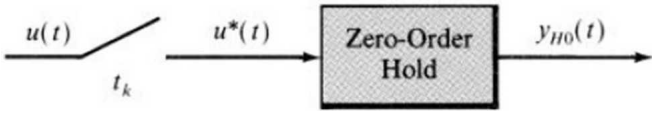
EXAMPLE 2.8. The input signal of an ideal sampler and a few samples of the output signal are illustrated in figure below. This type of signal is often called

a sampled-data signal.

Discrete-data signals u(tk) are often written more simply with the index k as the only argument, that is, u(k), and the sequence u(t1),u(t2),..., etc., becomes u(1), u(2),...,etc. Although sampling rates are in general non-uniform, however the previous example has a uniform sampling that is, tk + 1 — tk = T for all k. Definition 2.13: A hold, or data hold, device is one that converts the discrete time output of a sampler into a particular kind of continuous-time or analog signal.

EXAMPLE: A zero-order hold (or simple hold) is one that maintains (i.e., holds) the value of u(tk) constant until the next sampling time tk + 1 , as shown in the figure below. Note that the output y,H0(t) of the zero-order hold is continuous, except at the sampling times. This type of signal is called a piecewise-continuous signal.

zero-order hold switch block



Definition 2.14: An analog-to-digital (A/D) converter is a device that converts an analog or continuous signal into a discrete or digital signal.

Definition 2.15: A digital-to-analog (D/A) converter is a device that converts a discrete or digital signal into a continuous-time or analog signal.

EXAMPLE: Digital computers or microprocessors are often used to control continuous plants or processes. A/D and D/A converters are typically required in such applications, to convert signals from the plant to digital signals, and to convert the digital signal from the computer into a control signal for the analog plant. The joint operation of these elements is usually synchronized by a clock and the resulting controller is sometimes called a digital filter.

Computer Controlled Systems

Definition 2.16: A computer-controlled system includes a computer as the primary control element.

The most common computer-controlled systems have digital computers controlling analog or continuous processes. In this case, A/D and D/A converters are needed.

Definition 2.17: A transducer is a device that converts one energy form into another.

For example, one of the most common transducers in control systems applications is the potentiometer, which converts mechanical position into an electrical voltage.

Definition 2.18: The command v is an input signal, usually equal to the reference input r. But when the energy form of the command v is not the same as that of the primary feedback b, a transducer is required between the command v and the reference input r as shown.

Definition 2.19: When the feedback element consists of a transducer, and a transducer is required at the input, that part of the control system illustrated in below is called the error detector.

Definition 2.20: A stimulus, or test input, is any externally (exogenously) introduced input signal affecting the controlled output c. Note: The reference input r is an example of a stimulus, but it is not the only kind of stimulus.

Definition 2.21: A disturbance n (or noise input) is an undesired stimulus or input signal affecting the value of the controlled output c. It may enter the plant with u or m, or at the first summing point, or via another intermediate point.

Definition 2.22: The time response of a system, subsystem, or element is the output as a function of time, usually following application of a prescribed input under specified operating conditions.

Definition 2.23: A multivariable system is one with more than one input (multi input, MI-), more than one output (multi output, -MO), or both (multi

input-multi output, MIMO).

Definition 2.24: The term Controller in a feedback control system is often associated with the elements of the forward path, between the actuating (error) signal e and the control variable u. But it also sometimes includes the summing point, the feedback elements, or both, and some authors use the term controller and compensator synonymously. The context should eliminate ambiguity.

The following five definitions are examples of control laws, or control algorithms.

Definition 2.25: An on-off controller (two-position, binary controller) has

only two possible values at its output u, depending on the input e to the controller.

EXAMPLE: A binary controller may have an output u = +1 when the error signal is positive, that is e > 0, and u = -1 when e ≤ 0.

Definition 2.26: A proportional (P) controller has an output u proportional to its input e, that is, u = Kpe, where Kp is a proportionality constant.

Definition 2.27: A derivative (D) controller has an output proportional to the derivative of its input e, that is, u = KD de/dt, where KD is a proportionality constant.

Definition 2.28: An integral (I) controller has an output u proportional to the integral of its input e, that is, u = KI ∫ e(t) dt, where KI is a proportionality constant.

Definition 2.29: PD, PI, DI, and PID controllers are combinations of proportional (P), derivative (D), and integral (I) controllers. EXAMPLE: The output u of a PD controller has the form:

uPD = Kpe + KD de/dt

The output of a PID controller has the form:

uPID = Kpe + KD de/dt + KI ∫ e(t) dt

Servomechanisms

The specialized feedback control system called a servomechanism deserves special attention, due to its prevalence in industrial applications and control systems literature.

Definition 2.30: A servomechanism is a power-amplifying feedback control system in which the controlled variable c is mechanical position, or a time derivative of position such as velocity or acceleration.

Example: An automobile power-steering apparatus is a servomechanism. The command input is the angular position of the steering wheel. A small rotational torque applied to the steering wheel is amplified hydraulically, resulting in a force adequate to modify the output, the angular position of the front wheels.

Regulators Definition 2.31: A regulator or regulating system is a feedback control system in which the reference input or command is constant for long periods of time, often for the entire time interval during which the system is operational. Such an input is often called a setpoint.

A regulator differs from a servomechanism in that the primary function of a regulator is usually to maintain a constant controlled output, while that of a servomechanism is most often to cause the output of the system to follow a varying input.



Thank you!